

Platform Integral Predicts Slow-Light Window Formation in Photonic Crystal Waveguides

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Abstract

Slow-light phenomena in photonic crystal waveguides are typically associated with large group index values near band edges. However, the usable slow-light window is not determined solely by the peak group index.

In this work, we apply a platform integral framework to analyze slow-light window formation. Instead of relying on local extrema, we introduce a global measure based on the group velocity landscape:

$$P^*_{\text{slow}} = \int \exp(-|v_g|) \cdot \left| \frac{\partial v_g}{\partial \lambda} \right| d\lambda$$

where v_g is the group velocity. Using published data from photonic crystal waveguides exhibiting flatband slow modes, we show that usable slow-light windows correspond to extended regions of low and slowly varying group velocity, rather than isolated extrema.

This supports a structural interpretation: the peak group index determines the slow-light center, while the integrated weight of low-velocity plateaus governs the usable window width.

Together with previous polariton-based results, this suggests a cross-system platform integral mechanism for window

formation in hybrid mode systems.

1. Introduction

Slow-light effects in photonic crystal waveguides have been extensively studied due to their potential applications in optical buffering, enhanced nonlinear interactions, and sensing.

Conventional descriptions characterize slow-light behavior using the group index n_g , particularly its maximum value near band edges. However, experimental observations indicate that usable slow-light windows extend over finite bandwidths, often associated with flatband regions rather than sharp extrema.

This suggests that local extrema are insufficient to describe the full structure of slow-light windows.

2. Platform Integral Framework

We propose a global measure for slow-light window formation based on the group velocity landscape $v_g(\lambda)$:

$$P^*_{\text{slow}} = \int e^{\{-v_g/\langle v_g \rangle\}} \cdot e^{\{-|\partial_\lambda v_g|/\langle |\partial_\lambda v_g| \rangle\}} d\lambda$$

where the first factor weights low-velocity regions and the second favors flatness.

Using the relation:

$$v_g = \frac{c}{n_g}$$

the framework can also be expressed in terms of group index.

3. Empirical Consistency

Published results on photonic crystal waveguides demonstrate slow modes with nearly constant group index over finite bandwidths. For example, flatband slow-light modes exhibit approximately constant n_g over several nanometers, accompanied by high transmission efficiency and low dispersion.

These features correspond to extended low-velocity plateaus in the group velocity landscape, consistent with a large P^*_{slow} .

In contrast, configurations with sharp peaks in n_g but narrow bandwidths correspond to localized extrema with smaller integrated weight.

4. Core Result

We therefore propose:

The peak group index determines the slow-light center, while the integrated weight of low-velocity plateaus determines the usable slow-light window width.

This provides a structural alternative to extremum-based interpretations.

5. Cross-System Interpretation

This result aligns with polariton-based analysis, where stable windows are governed by the integral weight of low-residue plateaus:

$$Z_R(\theta) \rightarrow P^* \rightarrow Q_{\text{flat}}$$

Together, these results suggest a unified platform-based description of window formation across hybrid mode systems.

6. Conclusion

Slow-light window formation is governed by the global structure of the group velocity landscape rather than local extrema.

The platform integral provides a minimal and generalizable framework for describing this behavior, with potential applicability across multiple physical systems.

Keywords

slow light, photonic crystal waveguide, group index, platform integral, flatband, dispersion engineering, hybrid modes, stability window
